signal and image proccecing lab

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Lab 3

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separating mother and fetus ECG signal

In this experiment, you will work with the components of maternal and fetal heart signals (ECG). The maternal signal (Figure (i)), fetal signal (Figure (ii)), noise (Figure (iii)), and a mixture of the signals (Figure (iv)) have been drawn from 5-second windows of these signals. Using Blind Source Separation (BSS) methods, you will separate the mixed signals (Figure (iv)) into the original components (Figure (i)). Two methods, Principal Component Analysis (PCA) using Singular Value Decomposition (SVD) and Independent Component Analysis (ICA), will be employed in this experiment.

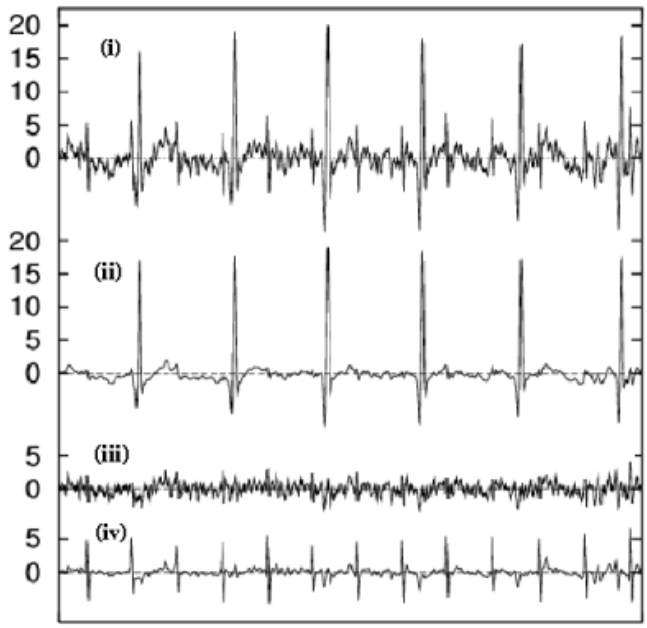

Figure 1. From top to bottom, the figures show: (i) Mixed maternal and fetal ECG signals. (ii) Maternal ECG signal. (iii) Noise. (iv) Fetal ECG signal.

In this experiment, you'll see that SVD, the first method, only uses second-order statistics, while ICA, the second method, leverages higher-order statistics. You will observe how the statistical distribution of the components affects the performance of these two methods in separating the signals.

In practical applications, one of the serious challenges for evaluating the performance of blind source separation methods is our lack of knowledge about the method of production and statistical characteristics of the sources. Therefore, in this experiment, you will use pre-recorded ECG and noise signals, and as a result, the quality of separation can be evaluated.

Initial Explanation about the Data

The maternal ECG, fetal ECG, and noise signals are available in the form of `noise1.dat`, `fecg1.dat`, and `mecg1.dat`. Using the `load` command, load them into the system. Each of these files contains data labeled as "recorded" in units of mV and sampled at a frequency of 256 Hz.

Combine these signals together to create a new mixed signal that closely resembles a real signal. Then, answer the following questions.

section 1

Plot the three signals: maternal, fetal, and noise, as well as the mixed observed signal, in the time domain.

solution

the output of the above task can be seen in below figure:

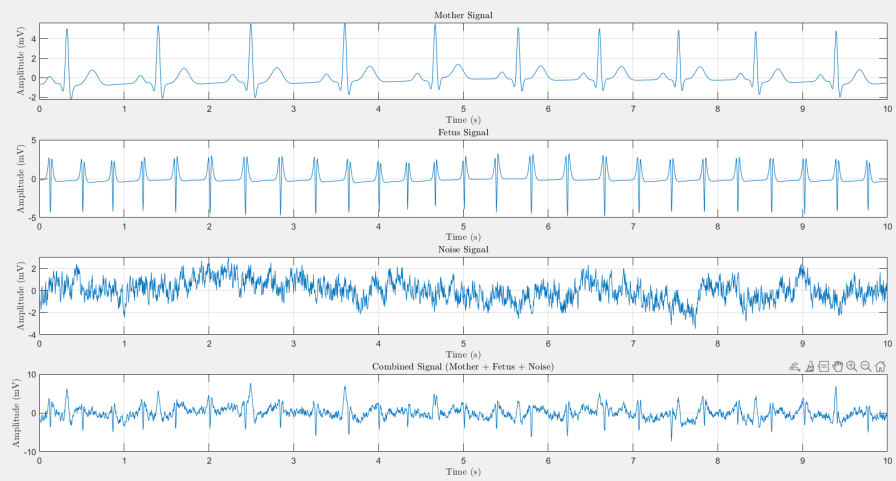

Figure 2. plot of maternal, fetal,noise and mixed signal

section 2

Plot the power spectrum of the three signals using the `pwelch` function. Are the frequency contents of the maternal and fetal signals comparable? Briefly mention the similarities and differences in these frequency contents.

solution

the plot of PSD of three signals is shown in figure 3.

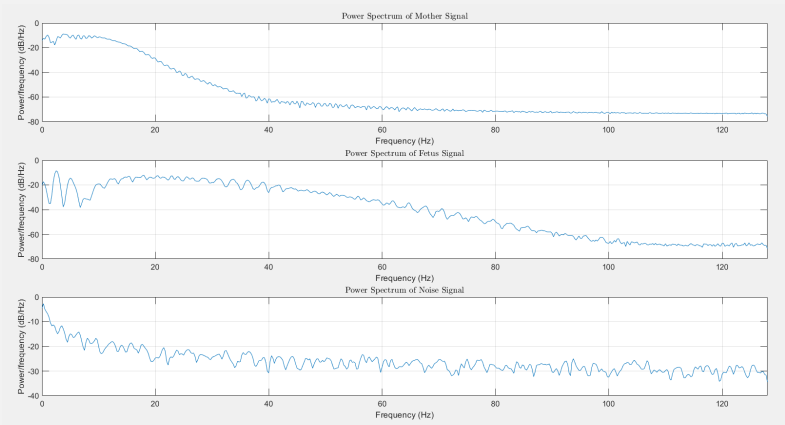

Figure 3. PSD of three signals

for similarities and differences we can say that: **Similarities:**

- **Frequency Range**: Both the maternal and fetal signals have significant power concentrated at lower frequencies, particularly in the range of 0–50 Hz. This indicates that both signals share similar low-frequency components.
- **Decay of Power with Frequency**: Both signals exhibit a general trend of decreasing power as frequency increases, especially above 50 Hz. This is typical in physiological signals, where lower frequencies often dominate.

Differences:

- **Peak Power Level**: The maternal signal has a higher peak power density at lower frequencies (around 0–10 Hz) compared to the fetal signal, indicating that the maternal signal is stronger in these low-frequency regions.
- **Variation in Frequency Content**: The fetal signal shows more oscillatory behavior in its frequency content, especially at higher frequencies, suggesting a more complex or variable composition. In contrast, the maternal signal has a smoother decay in its PSD.
- **Power Distribution in Mid-Range Frequencies**: The fetal signal retains more power up to around 60 Hz, whereas the maternal signal decays more rapidly after 20–30 Hz. This suggests that the fetal signal has richer content in the mid-range frequencies (20–60 Hz).

The maternal and fetal signals have comparable frequency ranges and general trends in power decay with increasing frequency. However, the maternal signal is more dominant at very low frequencies and has a smoother frequency distribution, while the fetal signal retains more power in mid-range frequencies and has a more complex pattern, indicating differences in their frequency contents.

section 3

Calculate and report the mean and variance of the three signals. Are these values comparable? Does the variance have a specific relationship with the frequency content of the signals?

solution

the mean and variances of these signals are shown in figure 4.

```
Mean of Mother Signal: -0.0000
Variance of Mother Signal: 1.0000
Mean of Fetus Signal: -0.0000
Variance of Fetus Signal: 1.0000
Mean of Noise Signal: -0.0000
Variance of Noise Signal: 1.0000
```

Figure 4. mean and variance of each signal

The means and variances of the three signals are comparable due to normalization. However, their variance does not reveal information about the frequency content. The PSD analysis is necessary to understand the distribution of power across frequencies, which varies significantly between the maternal, fetal, and noise signals.

section 4

Calculate the histogram of these signals using the `hist` command (to approximate the PDF of the components), and report the result. Also, calculate the 4th-order moment (kurtosis) of these signals using the `kurtosis` command, and report the result. Can you determine the degree of kurtosis in their PDFs based on these values?

solution

the histogram of the signals are calculated using `hist` function in matlab. the output is shown in figure 5.

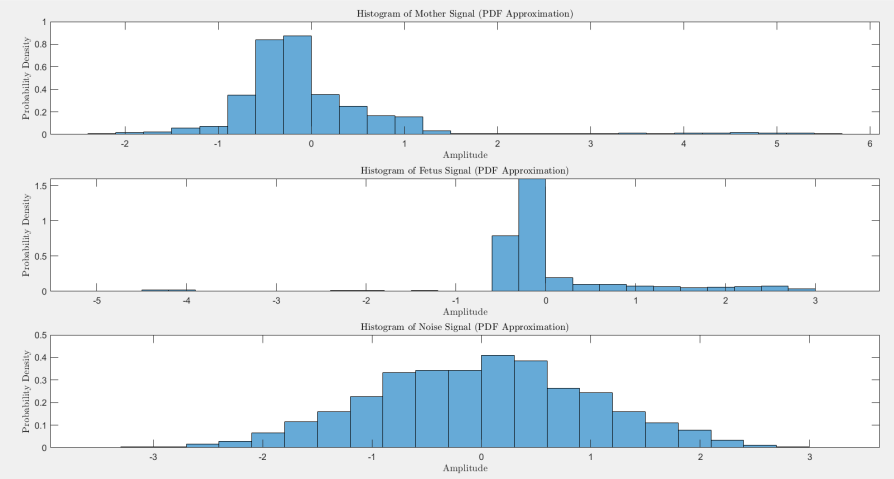

Figure 5. mean and variance of each signal

the kurtosis of the signals are as:

- Kurtosis of Mother Signal: 14.0421
- Kurtosis of Fetus Signal: 8.9901
- Kurtosis of Noise Signal: 2.7662

The histograms provide an estimate of the Probability Density Function (PDF) for each signal:

- **Mother Signal**: The histogram shows a high peak near zero with a skew towards negative values. This indicates a distribution that is centered close to zero but has more frequent occurrences of negative amplitudes than positive ones.

- **Fetus Signal**: The histogram is quite sparse, with most values close to zero and a few scattered negative amplitudes. This suggests a highly peaked distribution, which is less spread out compared to the mother signal.
- **Noise Signal**: The histogram of the noise signal shows a more symmetric, bell-shaped distribution centered around zero. This shape is indicative of a Gaussian or normal-like distribution, with values evenly distributed on both sides of zero.

the kurtosis values confirm that the mother and fetus signals have leptokurtic distributions with heavy tails, while the noise signal approximates a mesokurtic (Gaussian-like) distribution with lighter tails and a more uniform spread around the mean.

Signal Separation Using Singular Value Decomposition (SVD)

One of the standard and powerful techniques based on second-order statistics for filtering data is SVD. In this method, the data are projected onto orthogonal axes related to the highest variance in the data. These variances represent the most significant information extracted from the data (data-driven learning), and the initial measure of this variance is learned from the data.The variances are not adjustable. In this section, we want to analyze the content and variance of the components using SVD.

section 1

Read the data `X.dat` and display it using the command `plot3ch`, which has been provided to you. In this data, you do not have access to the hidden components (maternal, fetal signals, and noise). Instead, you have access to the mixed signal that has been recorded on three channels.The command `plot3ch` displays the data in two forms. In the first form, the three channels are shown in the time domain. In the second form, a scatter plot shows changes in one channel based on the other two channels. This provides more detailed information about the data (especially when the zero-mean condition has been applied). The points that are outside the central cloud are mostly related to the peaks of the QRS complexes.

Now, the goal is to use SVD to separate the maternal and fetal signals. Recall that SVD decomposes the matrix $Y \in \mathbb{R}^{M\times N}$, where the original data is represented as $Y = USV^T$. You can consider the matrix U as the decomposition of the signals, with the singular values $S \in \mathbb{R}^{M\times N}$ ordered along the diagonal. We aim to keep the first singular values and set the rest of the diagonal matrix to zero (this is shown in the eigenspectrum plot). By multiplying the matrices U, S, and V, the cleaned matrix is recovered as $C = YY^T$.

The matrix V contains the eigenvectors. After performing the analysis on the matrix, you can use the command `svd` to perform the decomposition of `X.dat` and retrieve the matrices U, V, and S.

solution

for this part first we use the following code to plot the `X.dat`:

```
function plot3ch(X,Fs,plot_title)
if nargin < 2
    Fs = 256;
end;
if nargin < 3
    plot_title = '3 Channel Data';
end;
[M,N] = size(X);
if N ~= 3;
    error('Input matrix must have 3 columns!');
end;
```

```
% Generate time-domain plot
%--------------------------
t = [1/Fs:1/Fs:M/Fs];
figure;
for i = 1:N
    subplot(N,1,i)
    plot(t,X(:,i)); ylabel(['Ch',int2str(i)]);
    axis([0 max(t) min(X(:,i))-abs(0.1*max(X(:,i))) max(X(:,i))+
        abs(0.1*max(X(:,i)))]);
end;
xlabel('Time (sec)');
subplot(N,1,1); title(plot_title);
figure;
plot3(X(:,1), X(:,2), X(:,3),'.m');
xlabel('Ch1'); ylabel('Ch2'); zlabel('Ch3');
title(plot_title);
grid on;
```

the output of this code is shown in figure 6 and 7.

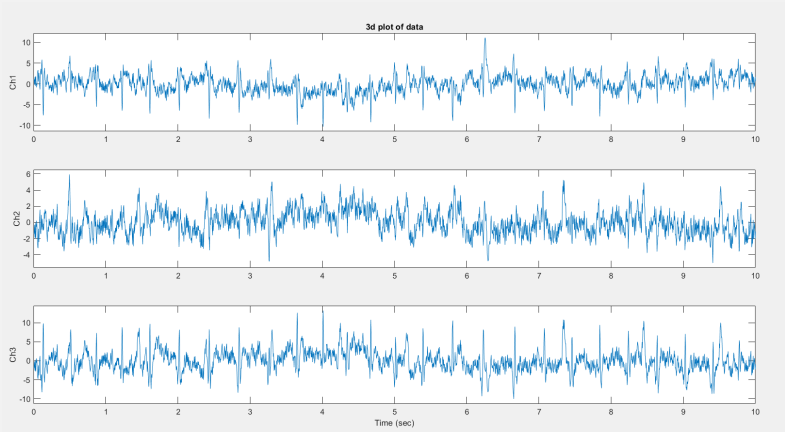

Figure 6. 3 channels of input signal

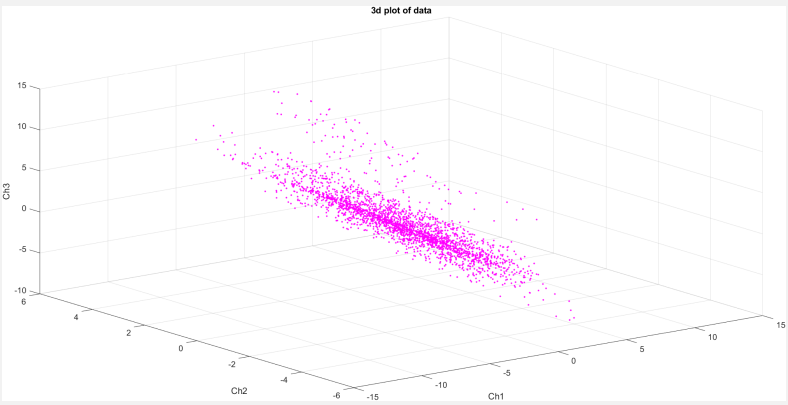

Figure 7. 3d plot of input signal

section 2

Plot the columns of matrix V using the `plot3dv` command, which has been provided to you. Pay attention to the following points:

(a) The `plot3dv` command receives matrix V and the corresponding singular values as input and performs normalization on them.

(b) Save the output image as `fig`, and also save the output matrices from the SVD.

(c) Remember that matrix $U \in \mathbb{R}^{M \times M}$ contains the original images projected onto the eigenvectors of matrix C. However, only the first N images are meaningful, as the image dimensions (M) cannot exceed the number of singular values (N).

solution

in this part we use this code:

```
function plot3dv(v, s, col)
% Input argument checking
%------------------------
if nargin < 2 | isempty(s)
    s = 100;
end;
if nargin < 3
    col = 'k';
end;
v = v(:); % ensure that eigenvector is in column format
[m, n] = size(v);
if (n ~= 1 | m ~= 3)
    error('vector must be 3x1')
end;
if s == 1  % legacy code: does not affect function
    ln = 1/sqrt((v(1)*v(1))+(v(2)*v(2))+(v(3)*v(3)));
end;

% Plot eigenvector on 3D plot
%----------------------------
sn = sqrt(s);
hold on;
plot3(sn*[-1*v(1) v(1)],sn*[-1*v(2) v(2)],sn*[-1*v(3) v(3)],col);
grid on;
view([1,1,1])
```

the output of this code for V of the input matrix is shown in figure 8.

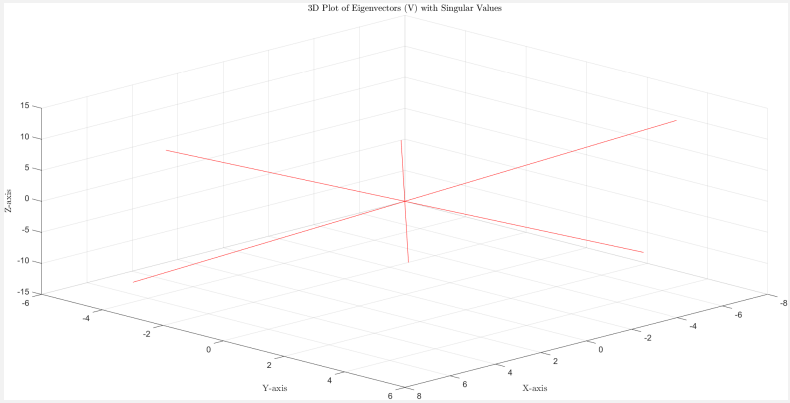

Figure 8. output of V vector of SVD of input data

section 3

Plot the first three columns of matrix U. For each of these images, determine which feature (maternal, fetal, noise, or mixture) is dominant. Which of these images is more helpful in distinguishing the fetal signal? Also, plot the eigenspectrum using the `stem` command and include this in your report.

It is worth mentioning that this type of projection is not clinically meaningful. Traditional diagnoses of ECG signals are done by checking for high and low thresholds in the ECG images. However, by using the SVD decomposition, we are moving into a non-clinical space where specific thresholds for diagnosing the fetal ECG are no longer meaningful. Therefore, if we want to retrieve clinical information after performing the SVD, we need to invert the data back into the original space using the inverse SVD. Then, we need to correct the fetal-related singular values and discard the rest, and reconstruct the sensor space by modifying matrix S accordingly and multiplying it back with the same matrices U and V. This way, we can return to the fetal sensor space (by taking the fetal components) after performing the SVD.

solution

the outputs are shown in figure 9 and 10.

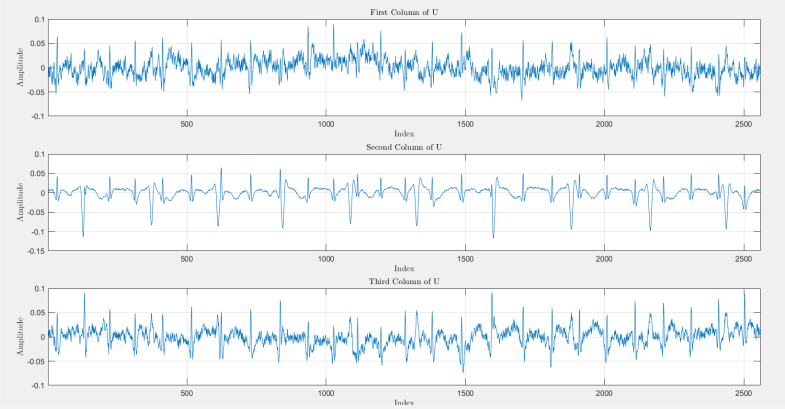

Figure 9. plot of first 3 columns of U

as it can be seen in above figure the second image is more helpful in distinguishing the fetal signal as it is more similar to the fetal signal. the eignspectrum of the U vector is shown in figure 10.

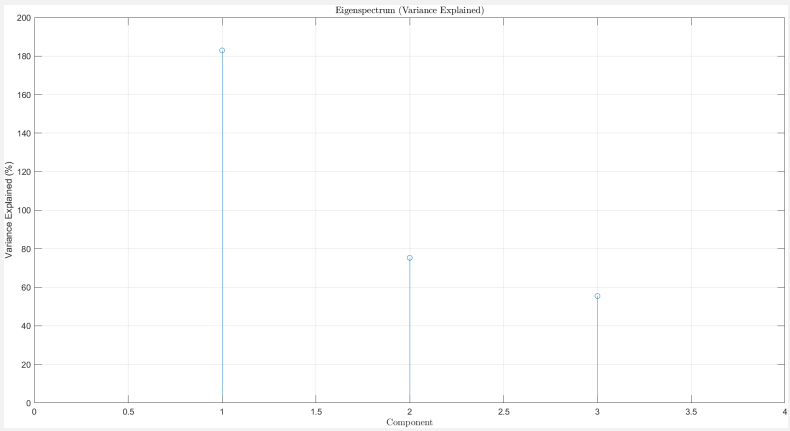

Figure 10. eignspectrum of U vector

section 4

Display the three reconstructed fetal sensor channels and describe the effect of SVD decomposition and the applied corrections. Was the reconstruction of the fetal signal successful?

solution

by reconstructing the fetal signal we reach the reconstruction error of 191.1528. but as it can be seen we couldn't separate the channels using SVD method.

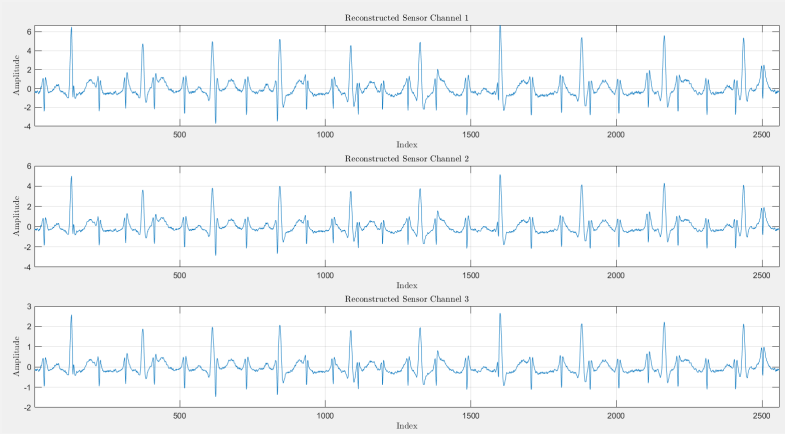

Figure 11. reconstruction of input signal using SVD

Source Separation Using ICA

In this section, we use a higher-order statistics-based technique called Independent Component Analysis (ICA) for source separation of mixed signals, such as brain and noise signals. The mixed pattern can be represented as a "mixed matrix" $\mathbf{X}$ and, more precisely, can be expressed mathematically as:

$$\mathbf{X}^T = \mathbf{A}\mathbf{Z}^T$$

It is worth mentioning that, to represent the observation matrix $\mathbf{X}$ in the form of $channels \times timesamples$, the relationship on both sides has been transposed. Here, N represents the number of channels, and M represents the number of time samples ($\mathbf{X} \in \mathbb{R}^{M \times N}$).

Therefore, the estimated sources $\mathbf{Z}$ can be written as follows:

$$\hat{\mathbf{Z}}^T = \mathbf{W}\mathbf{X}^T$$

where $\mathbf{W}$ is the separation matrix, and $\mathbf{W}^{-1}$ is an approximation of the mixing matrix $\mathbf{A}$.

section 1

Apply the ICA technique on the observation matrix $\mathbf{X}$ ($\mathbf{X}^T$ in the code file provided to you), compute the matrix $\mathbf{W}$ and save it, then compute the estimated source matrix $\hat{\mathbf{Z}}$ from $\mathbf{W}^{-1}$ and save it as well.

solution

Plot data:

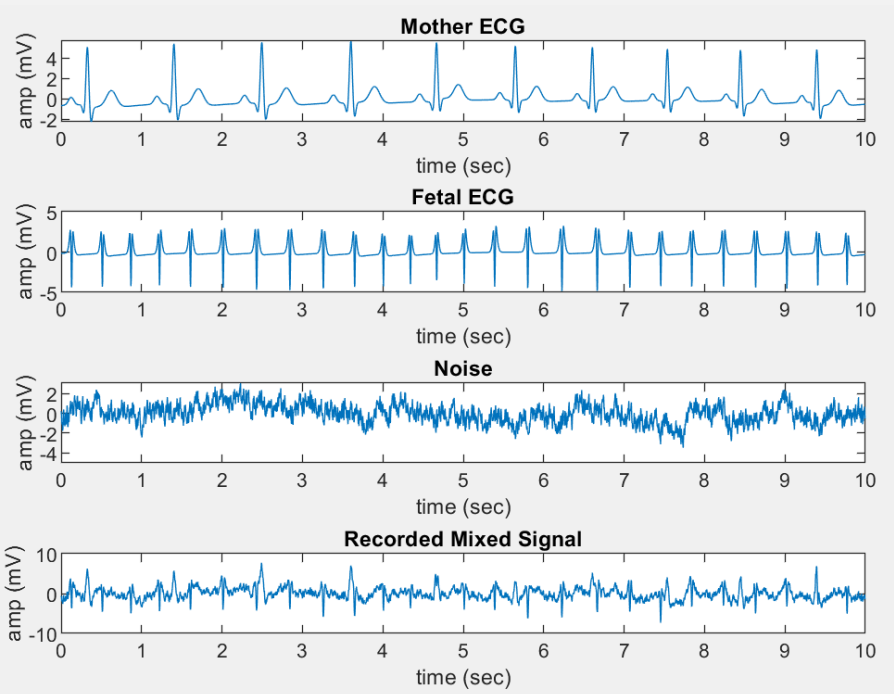

Figure 12. plot

Part of the code for saving matrices that the question asked for:

```
[W, Z_hat] = ica(X');

A = inv(W);
```

Figure 13. code for saving matrices

section 2

Plot the initial data as a scatter plot using the `plot3ch` function (a function with three components), and use `plot3dv` to plot the rows of $\mathbf{W}^{-1}$. The `plot3dv` function takes the matrix $\mathbf{W}^{-1}$ as input and can retrieve and display multiple forms, saving the figure in `.fig` format.

solution

Plot channels:

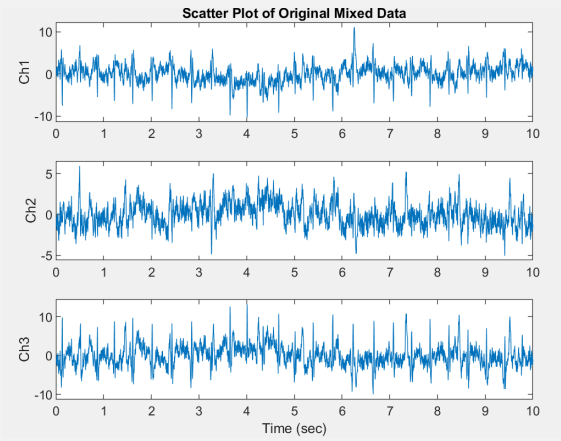

Figure 14. Channels

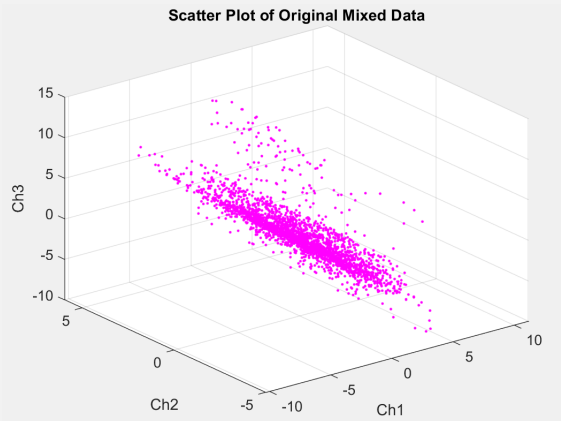

Figure 15. Scatter plot

Visualization of W inverse columns:

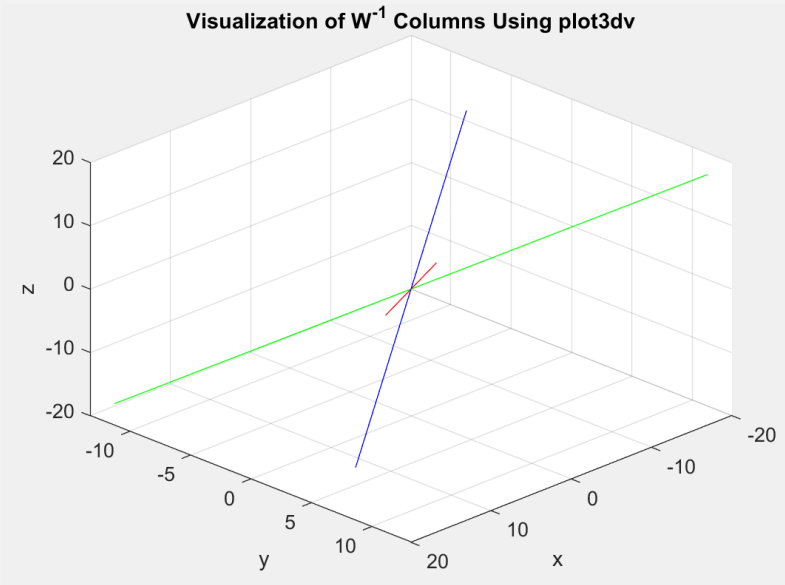

Figure 16. Visualization of W inverse columns

section 3

Plot three columns of the matrix $\hat{\mathbf{Z}}^T$ as equivalent to three rows of the matrix $\mathbf{W}^{-1}$. Choose one of the columns that best represents the genetic component, keeping it as is, and set the remaining columns to zero. Use the corrected matrix $\hat{\mathbf{Z}}$ and matrix $\mathbf{W}^{-1}$ to map the genetic component back to the ECG sensor space.

solution

Visualization of A columns:

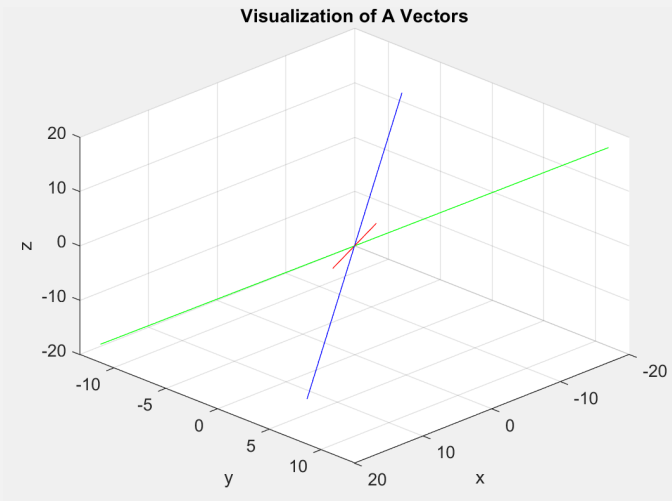

Figure 17. Visualization of A columns

section 4

Plot and save the data reconstructed with ICA. In your opinion, has the genetic component been successfully reconstructed?

solution

Plot of separated sources:

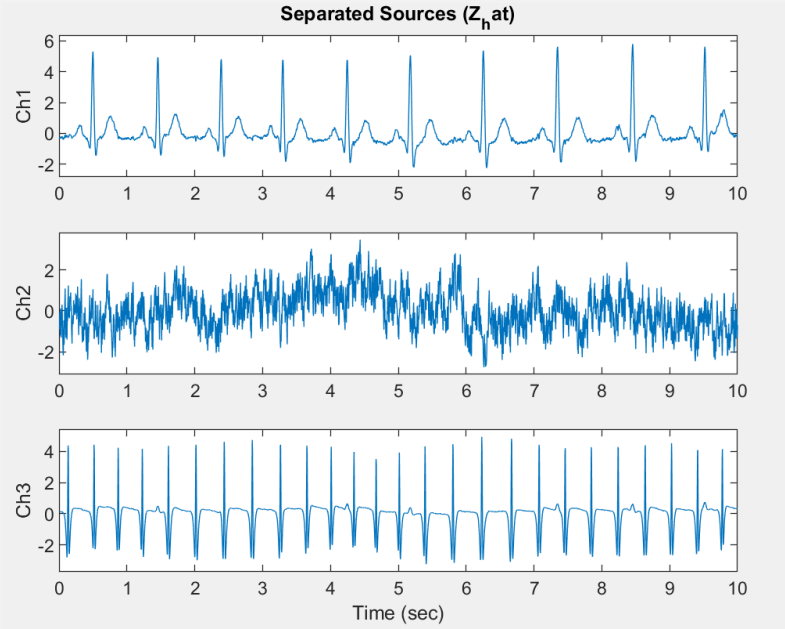

Figure 18. Plots

Scatter plot of separated sources:

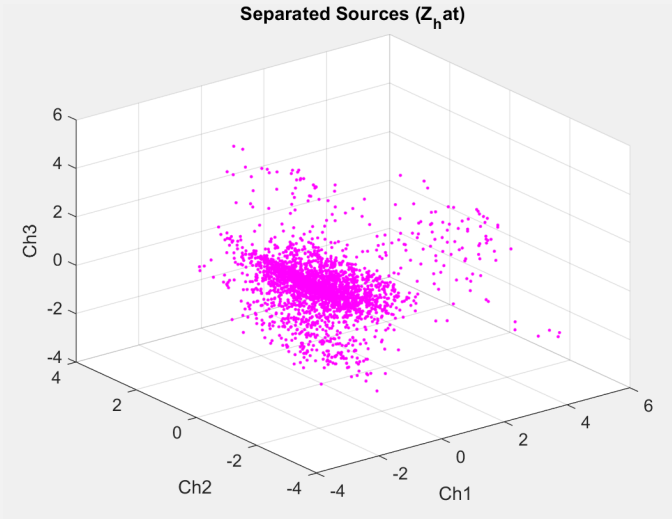

Figure 19. Scatter plot

Reconstructed signal

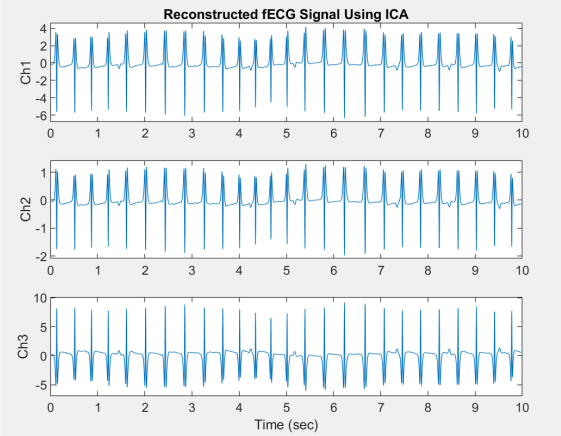

Figure 20. Reconstructed signal

As we can see from the plot above and compared with the original signal, it's clear that the reconstruction was been successful.

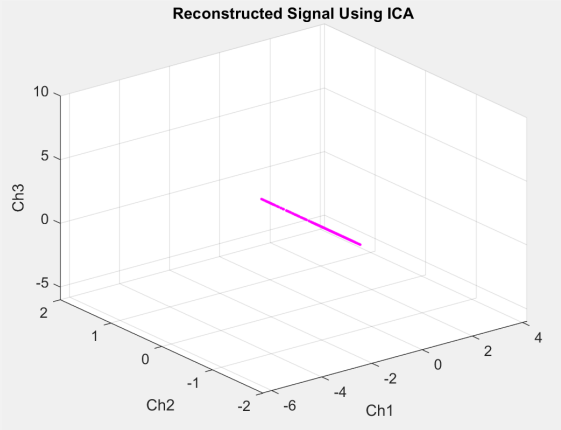

Figure 21. Visualization of A columns

The ICA reconstruction was successful in isolating the fetal ECG component from the mixed signal. As ICA leverages statistical independence, it can better separate components with unique non-Gaussian distributions. The reconstructed fetal signal aligns well with the original in both waveform and amplitude, confirming that ICA effectively isolated the genetic component, even with overlapping maternal and noise signals.

Comparisons

section 1

Plot the following graphs in a 3D space (corresponding to three channels): - The scatter plot of the observation matrix **X**. - The scatter plot of the reconstructed observation matrix obtained in parts two and three. - The direction of the columns of matrix **V**. - The direction of the columns of matrix $\mathbf{W}^{-1}$.

Compare the graphs and analyze the results obtained from both methods. Compute and compare the angles between different directions using the `dot` command to calculate various dot products between the axes.

The main question here is how successful this method has been in separating the sources. Has it only managed to separate cases where the knowledge about the sources and noise was complete? Therefore, given that synthetic data was used here with knowledge about the desired signal (fECG) and noise (mECG+noise) being complete, it's clear that the initial matrix **X** includes the ideal signal fECG in `fecg1.dat` and the scaled version of `fecg2.dat` that has been provided to you as `fecg3.dat`.

solution

plot of original observation matrix:

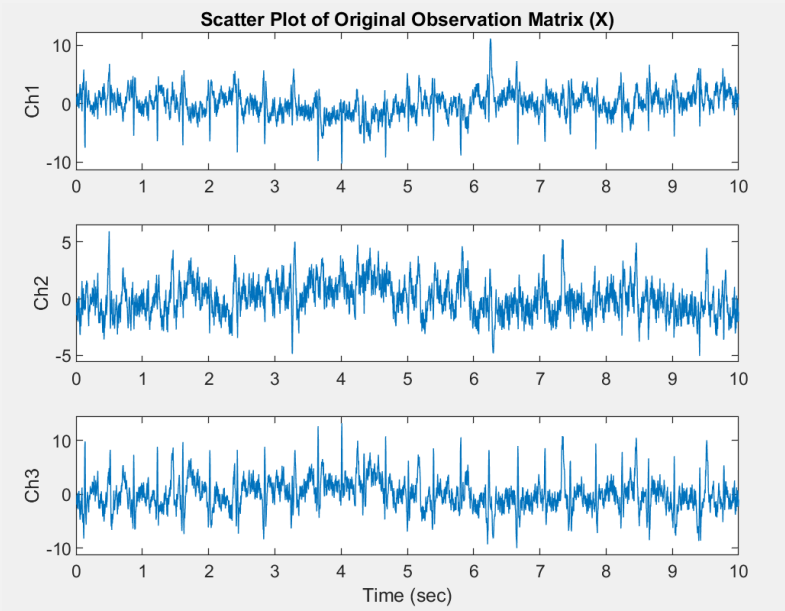

Figure 22. original observation matrix

Scatter plot of original observation matrix:

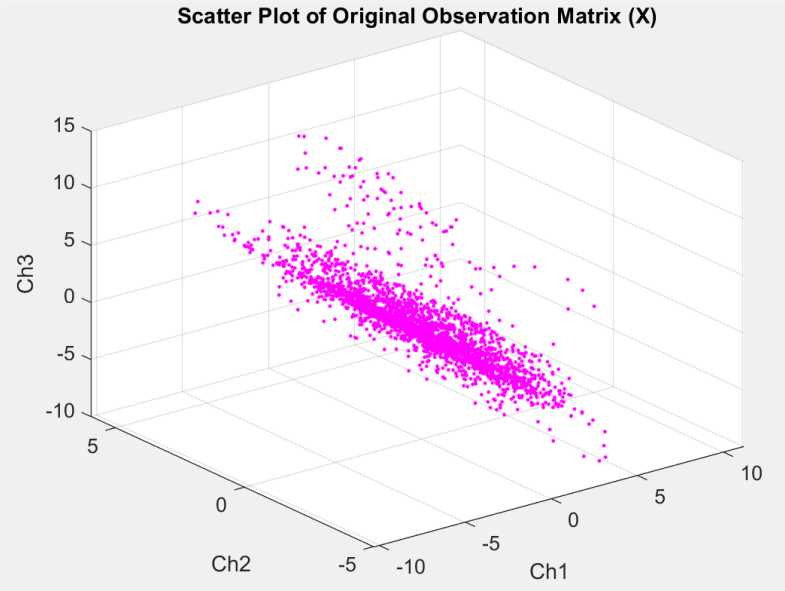

Figure 23. Scatter plot

Reconstructed with ICA:

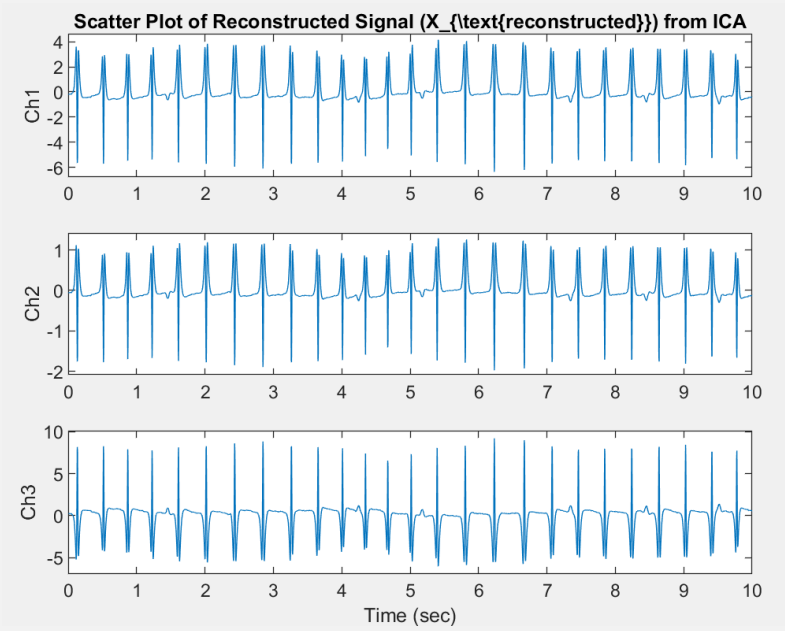

Figure 24. plot

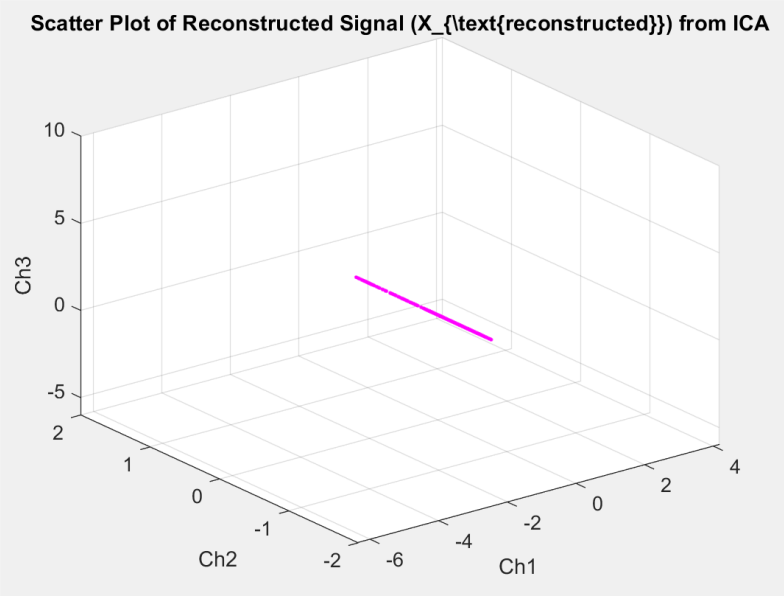

Figure 25. Scatter plot

Reconstructed with SVD:

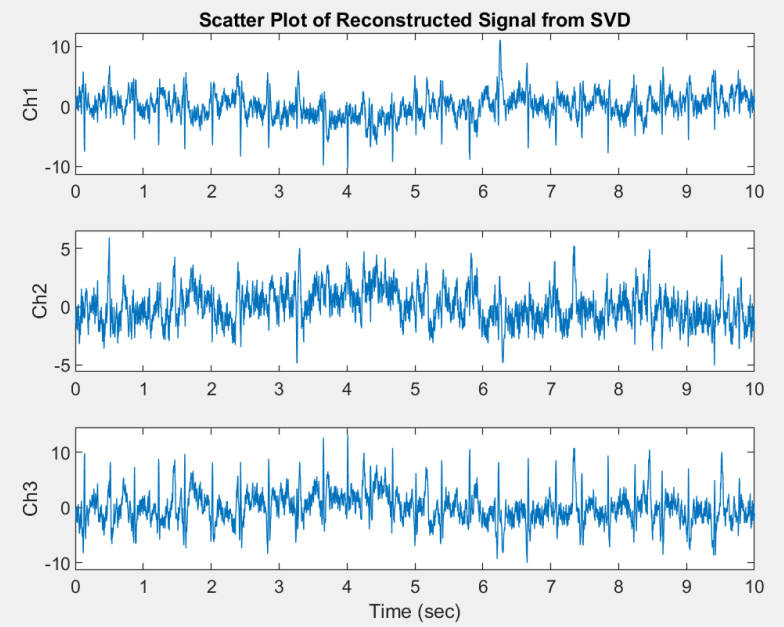

Figure 26. plot

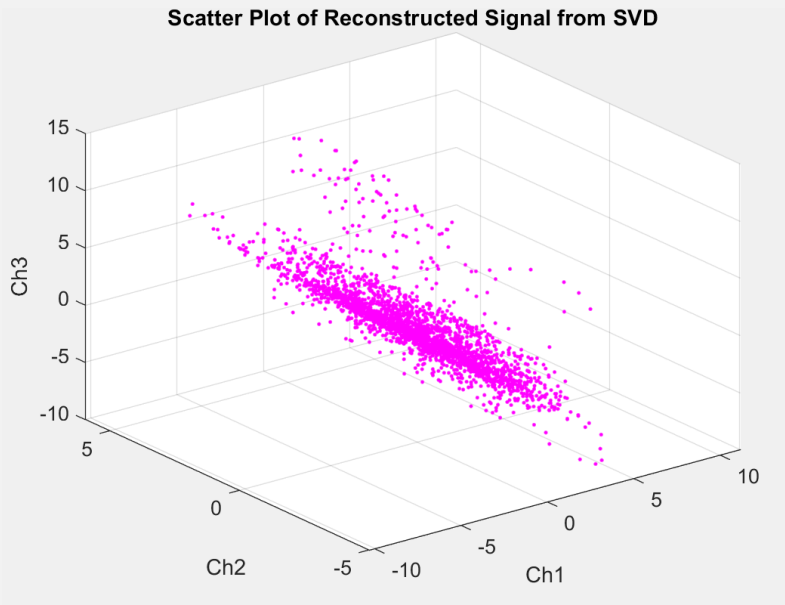

Figure 27. Scatter plot

Visualization W inverse columns:

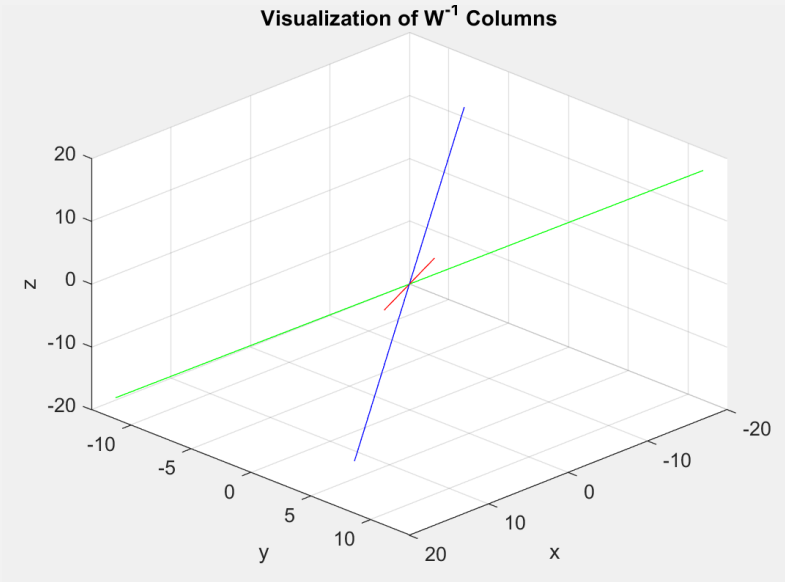

Figure 28. Visualization W inverse columns

Visualization of V columns:

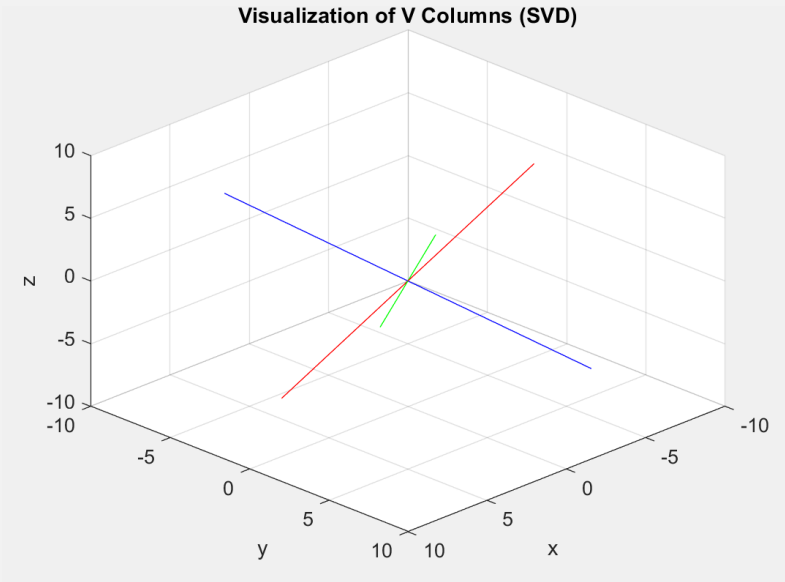

Figure 29. Visualization of V columns

Comparison of the ICA reconstructed with truth signal:

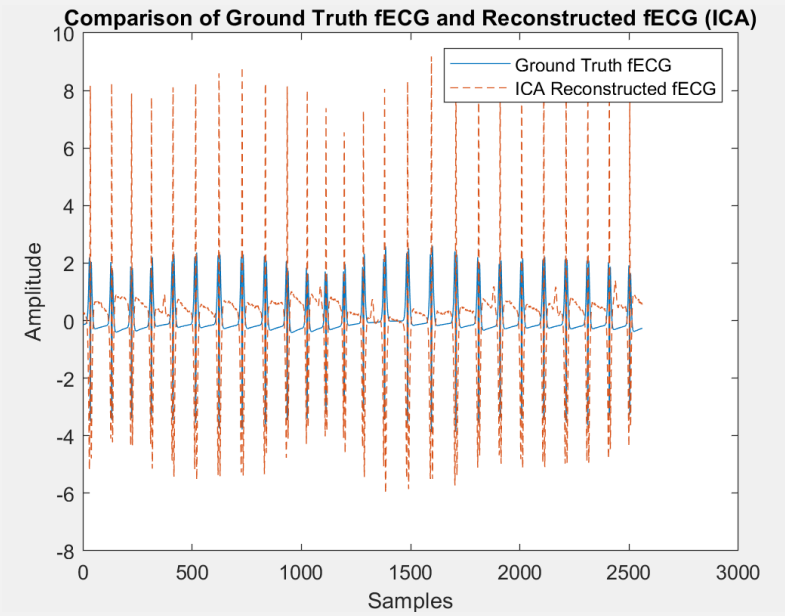

Figure 30. Comparison of the ICA reconstructed with truth signal

Comparison of the SVD reconstructed with truth signal

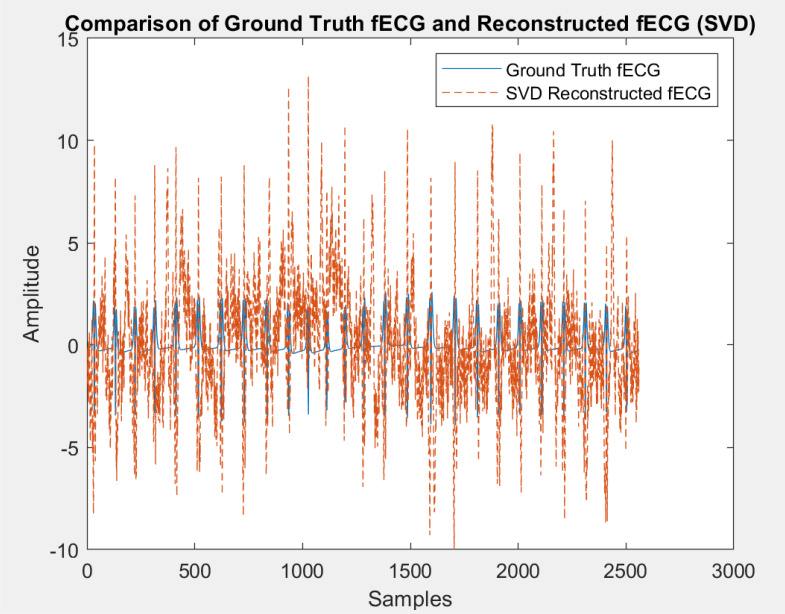

Figure 31. Comparison of the SVD reconstructed with truth signal

Angles:

```
Angle between v1 and w1: 62.1534 degrees
Angle between v2 and w2: 60.9584 degrees
Angle between v3 and w3: 38.4846 degrees
Norm of v1: 1.6350
Norm of v2: 2.9135
Norm of v3: 2.3079
Norm of w1: 1.0030
Norm of w2: 1.0003
Norm of w3: 1.0030
```

Figure 32. Angles and norms

Vector V is an orthogonal and unity norm.
The columns of matrix V (SVD) represent orthogonal directions capturing maximum variance, primarily driven by the maternal signal due to its larger amplitude. In contrast, the columns of W^{-1} (from ICA) are oriented based on statistical independence rather than variance. This alignment reflects ICA's focus on separating signals with unique distributional properties, which explains why ICA was more successful in isolating the fetal ECG component, whereas SVD aligned more with the dominant maternal component.

section 2

Plot the reconstructed ideal signal from both ICA and SVD methods (one of the signal channels that was reconstructed by both methods) and compare their performance relative to the ideal signal `fecg2.dat` to determine which method is superior.

solution

Comparing reconstructed signals from ICA and SVD methods with the truth signal:

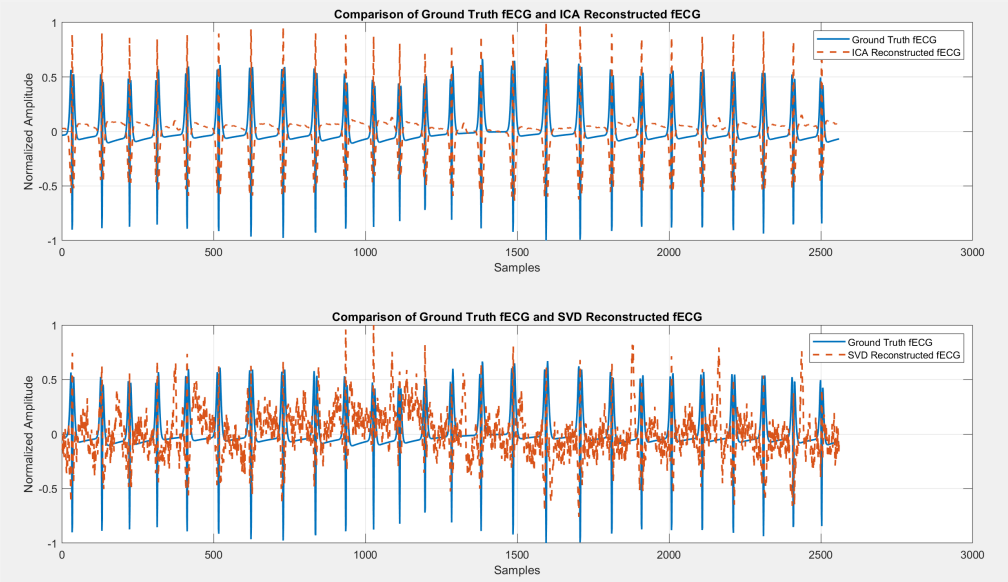

Figure 33. Compare signals

Compare reconstruction errors:

```
Reconstruction Error for ICA: 20.7112
Reconstruction Error for SVD: 787.3829
```

Figure 34. Compare errors

As we can see in the results above, ICA method performs much better than SVD in reconstruction.The reconstructed signal from ICA is more similar to the original signal than SVD. Also, the reconstruction error for ICA is much smaller than the ICA.

section 3

Finally, calculate the correlation coefficient between the ideal signal `fecg2` and the reconstructed signal from each of the two methods, by comparing one of the signal channels that was reconstructed using the `corcoef` command.

solution

Comparing correlations:

```
Correlation coefficient between ground truth and ICA reconstructed fECG: -0.9978
Correlation coefficient between ground truth and SVD reconstructed fECG: -0.5828
```

Figure 35. Comparing correlations

As it's clear in the results above, we can see that the correlation of the reconstructed signal from ICA with the truth signal is much more than the correlation of the reconstructed signal from SVD.
The superior performance of ICA in terms of both reconstruction error and correlation coefficient stems from its ability to leverage higher-order statistics, which allows it to separate non-Gaussian sources effectively. The fetal ECG, having unique higher-order statistical properties distinct from the maternal ECG and noise, aligns well with ICA's assumptions. SVD, limited by second-order statistics, cannot exploit these higher-order differences, leading to less accurate separation and higher reconstruction errors.

section 4

Perform a final comparison between the ICA and SVD methods and discuss the advantages and disadvantages of each method in a general form.

solution

- **Advantages of ICA:** ICA leverages higher-order statistics, making it effective for separating non-Gaussian signals, such as the fetal ECG in this experiment. The ICA method successfully reconstructed the fetal signal with higher accuracy, as seen in both the correlation coefficient and the lower reconstruction error compared to SVD. ICA's performance was superior in capturing the true components, particularly in cases where source independence is an underlying assumption.
- **Disadvantages of ICA:** One limitation of ICA is its reliance on the assumption of statistical independence between sources, which may not always be realistic in complex, real-world signals. Additionally, ICA is more computationally intensive than SVD, which may be a consideration in applications requiring real-time processing.
- **Advantages of SVD:** SVD is computationally simpler and requires only second-order statistics, which makes it faster and effective for applications where less signal separation is needed or where computational resources are limited. In the experiment, SVD showed acceptable separation, though it was less accurate than ICA.
- **Disadvantages of SVD:** Since SVD only leverages second-order statistics, it may be insufficient for separating signals with overlapping frequency components or non-Gaussian characteristics, as observed in this experiment with the fetal ECG signal. Consequently, SVD had a higher reconstruction error and lower correlation with the original signal than ICA.

Overall, the results indicate that ICA is preferable for tasks requiring detailed source separation when signals are non-Gaussian and statistically independent, while SVD remains a viable option for simpler, less specific separation tasks or when computational resources are limited.

section 5

State the most important insights you have learned from this experiment.

solution

- **Effectiveness of Higher-Order Statistics in Signal Separation:** The results showed that higher-order statistics, as utilized by ICA, are highly effective in separating mixed signals with non-Gaussian distributions. This underscores the advantage of ICA in applications involving complex, independent sources.
- **Limitations of SVD for Source Separation:** Although SVD is useful for dimensionality reduction and simple separation tasks, it may fall short in applications requiring fine-grained signal separation. Its reliance on second-order statistics limits its effectiveness when components share similar frequency characteristics.
- **Importance of Prior Knowledge on Signal Characteristics:** The success of separation methods depends significantly on understanding the statistical properties of the signals. ICA performed well because the signals were non-Gaussian and independent, aligning with ICA's assumptions. This suggests that selecting the right method requires careful consideration of signal characteristics.
- **Practical Applications of Blind Source Separation (BSS):** This experiment highlights the potential of BSS methods like ICA and SVD in practical scenarios, such as medical signal processing. Understanding the strengths and limitations of each method can guide their application in separating biologically meaningful components.